Topic: Homogeneous equations

Question: Find the solution to the differential equation.

$$(x+y)y' = x - y$$

Answer choices:

A $C = x^2 + 2xy + y^2$

B $C = \dfrac{1}{x^2 + 2xy + y^2}$

C $C = x^2 - 2xy - y^2$

D $C = 1 - 2xy - y^2$

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Solution: C

We start by solving for y',

$$y' = \frac{x - y}{x + y}$$

then we'll multiply through the numerator and denominator on the right side by $1/x$.

$$y' = \frac{x - y}{x + y} \cdot \frac{\frac{1}{x}}{\frac{1}{x}}$$

$$y' = \frac{x\left(\frac{1}{x}\right) - y\left(\frac{1}{x}\right)}{x\left(\frac{1}{x}\right) + y\left(\frac{1}{x}\right)}$$

$$y' = \frac{1 - \frac{y}{x}}{1 + \frac{y}{x}}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = \frac{1 - v}{1 + v}$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = \frac{1 - v}{1 + v} - v$$

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$$x\frac{dv}{dx} = \frac{1-2v-v^2}{1+v}$$

$$\frac{1+v}{1-2v-v^2}\,dv = \frac{1}{x}\,dx$$

and then integrate both sides, using a substitution for the integral on the left, with $u = 1-2v-v^2$ and $du = (-2-2v)\,dv$, and $dv = (1/(-2-2v))\,du$.

$$\int \frac{1+v}{1-2v-v^2}\,dv = \int \frac{1}{x}\,dx$$

$$\int \frac{1+v}{u}\left(\frac{1}{-2-2v}\,du\right) = \int \frac{1}{x}\,dx$$

$$-\frac{1}{2}\int \frac{1+v}{u}\left(\frac{1}{1+v}\,du\right) = \int \frac{1}{x}\,dx$$

$$-\frac{1}{2}\int \frac{1}{u}\,du = \int \frac{1}{x}\,dx$$

$$-\frac{1}{2}\ln u = \ln x + C$$

$$\ln u^{-\frac{1}{2}} = \ln x + C$$

$$e^{\ln u^{-\frac{1}{2}}} = e^{\ln x + C}$$

$$u^{-\frac{1}{2}} = Cx$$

Back-substitute for u.

$$(1-2v-v^2)^{-\frac{1}{2}} = Cx$$

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$$\frac{1}{1-2v-v^2} = Cx^2$$

$$1 = Cx^2(1-2v-v^2)$$

Back-substitute for v.

$$1 = Cx^2\left(1-2\frac{y}{x}-\frac{y^2}{x^2}\right)$$

$$1 = C(x^2-2xy-y^2)$$

$$C = x^2-2xy-y^2$$

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Topic: Homogeneous equations

Question: Find the solution to the differential equation.

$$x^3y' = x^2y + y^3$$

Answer choices:

A $$y = \pm \frac{x}{\sqrt{C - 2\ln x}}$$

B $$y = \frac{1}{\sqrt{C - 2\ln x}}$$

C $$y = \frac{x}{\sqrt{C - 2\ln x}}$$

D $$y = \pm \frac{x}{\sqrt{C - \ln x}}$$

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Solution: A

To put the equation in standard form, we'll divide through by x^3.

$$\frac{x^3}{x^3}y' = \frac{x^2y}{x^3} + \frac{y^3}{x^3}$$

$$y' = \frac{y}{x} + \left(\frac{y}{x}\right)^3$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = v + v^3$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = v^3$$

$$x\frac{dv}{dx} = v^3$$

$$\frac{1}{v^3}\,dv = \frac{1}{x}\,dx$$

and then integrate both sides.

$$\int \frac{1}{v^3}\,dv = \int \frac{1}{x}\,dx$$

$$-\frac{1}{2v^2} = \ln x + C$$

$$\frac{1}{v^2} = -2\ln x + C$$

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$$v^2 = \frac{1}{C - 2\ln x}$$

$$v = \pm \frac{1}{\sqrt{C - 2\ln x}}$$

Back-substitute for v.

$$\frac{y}{x} = \pm \frac{1}{\sqrt{C - 2\ln x}}$$

$$y = \pm \frac{x}{\sqrt{C - 2\ln x}}$$

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Topic: Homogeneous equations

Question: Find the solution to the differential equation.

$$2xy' = 2y + 4\sqrt{xy}$$

Answer choices:

A $y = x\sqrt{\ln x + C}$

B $y = x(\ln x + C)^2$

C $y = \sqrt{\ln x + C}$

D $y = (\ln x + C)^2$

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Solution: B

To put the equation in standard form, we'll divide through by $2x$.

$$\frac{2x}{2x}y' = \frac{2y}{2x} + \frac{4\sqrt{xy}}{2x}$$

$$y' = \frac{y}{x} + 2\sqrt{\frac{y}{x}}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = v + 2\sqrt{v}$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = 2\sqrt{v}$$

$$x\frac{dv}{dx} = 2\sqrt{v}$$

$$\frac{1}{2\sqrt{v}}\,dv = \frac{1}{x}\,dx$$

and then integrate both sides.

$$\int \frac{1}{2\sqrt{v}}\,dv = \int \frac{1}{x}\,dx$$

$$\sqrt{v} = \ln x + C$$

$$v = (\ln x + C)^2$$

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Back-substitute for v.

$$\frac{y}{x} = (\ln x + C)^2$$

$$y = x(\ln x + C)^2$$

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